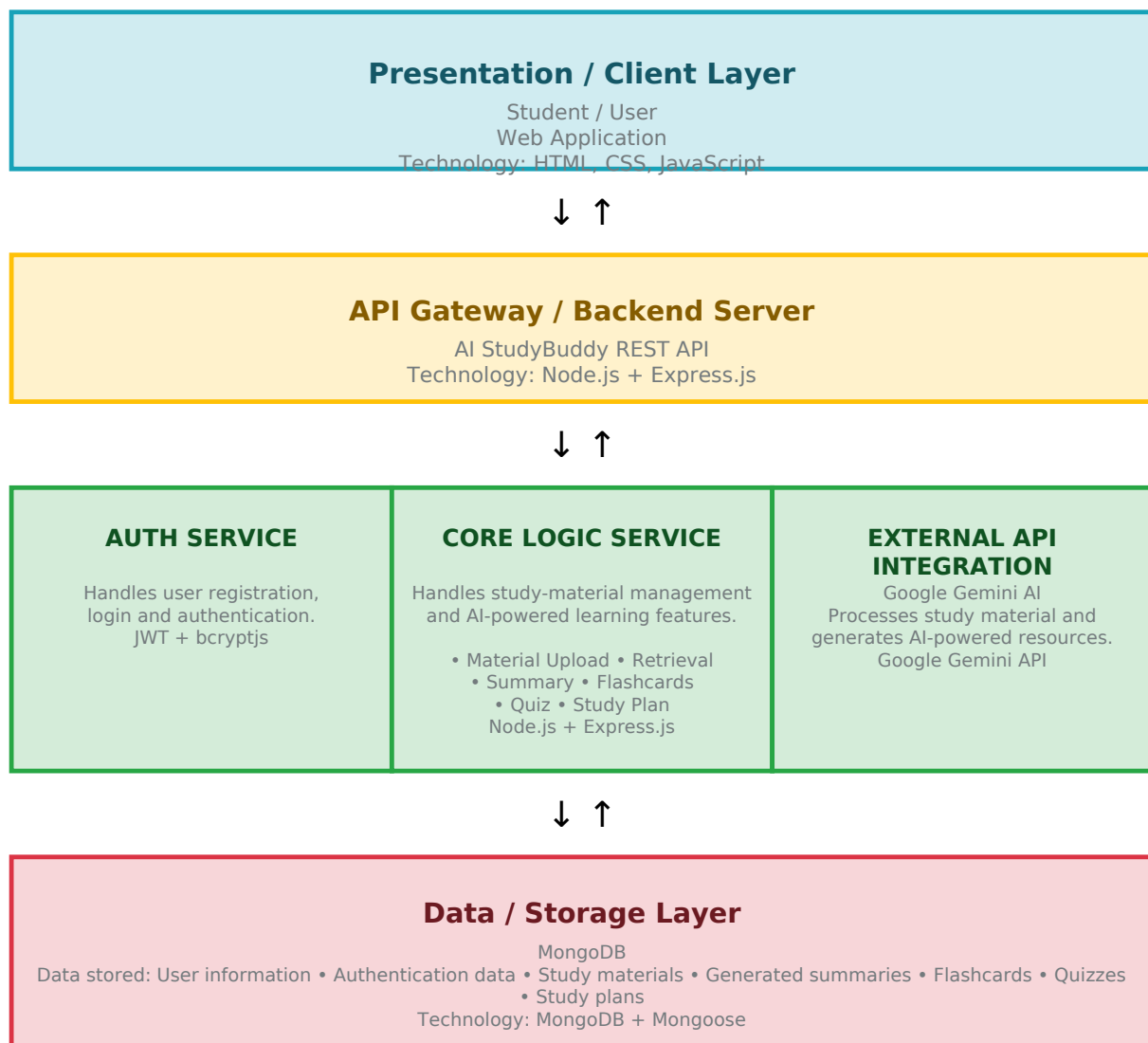


Solution Architecture

Date	08 October 2026
Team ID	UNM1611Firefox
Project Name	AI StudyBuddy API
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
1. PRESENTATION LAYER	Provides the user interface through which students can register, log in, upload study materials, view generated content, and interact with the StudyBuddy system.	HTML, CSS, JavaScript
2. API GATEWAY / BACKEND	Receives client requests, routes them to the appropriate backend functionality, handles API responses, and connects the application with the database and AI service.	Node.js, Express.js
3. AUTH SERVICE	Handles user registration and login and protects API endpoints using JWT authentication. Passwords are securely hashed before storage.	JWT, bcryptjs
4. CORE LOGIC SERVICE	Manages study materials and controls the AI-powered learning workflow including summary, flashcard, quiz, and study-plan generation.	Node.js, Express.js
5. EXTERNAL AI SERVICE	Processes study-material content and generates AI-powered learning resources based on user requests.	Google Gemini API
6. DATABASE / DATA STORAGE	Stores user information, study materials, authentication-related data, and generated learning resources.	MongoDB, Mongoose